

ISHMAL KHAN

Mathematics Undergraduate | Data Analyst | Financial Modeling Enthusiast

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PROFESSIONAL SUMMARY

Mathematics undergraduate (8th Semester) at Bahauddin Zakariya University (BZU) with strong expertise in data analysis, financial modeling, and quantitative methods. Proficient in Python, SQL, R, and MATLAB with hands-on experience in machine learning, statistical analysis, time-series forecasting, and financial valuation. Passionate about applying mathematical foundations to real-world data science and finance problems.

EDUCATION

Bachelor of Science in Mathematics

Bahauddin Zakariya University (BZU) | In Progress — 8th Semester

Relevant Coursework: Calculus, Algebra, Functional Analysis, Discrete Mathematics, Ordinary & Partial Differential Equations, Stochastic Calculus, Complex Analysis, Operations Research, Optimization Theory, Portfolio Theory & Risk Management, Machine Learning, Time Series Analysis, Actuarial Mathematics, Introduction to Deep Learning, Foundations and Applications of Generative AI, Business Administration, Strategic Management, Entrepreneurship

TECHNICAL SKILLS

Programming Languages: Python (Expert), SQL (Advanced), R, MATLAB

Data & Visualization: Pandas, NumPy, Matplotlib, Seaborn, Plotly

Machine Learning & Stats: Predictive Modeling, Statistical Analysis, Hypothesis Testing, Data Mining

Financial Analysis: Financial Modeling, Portfolio Analysis, Risk Management, DCF Valuation, Time-Series Analysis (ARIMA, Prophet)

Tools & Software: Jupyter Notebook, Microsoft Excel, Power BI, R Studio, AutoCAD, PowerPoint, Canva

PROJECTS

Stock Price Forecasting

- Developed time-series forecasting models using ARIMA and Facebook Prophet algorithms
- Created interactive visualizations using Plotly to present forecast results and confidence intervals

Financial Statement Valuation

- Performed comprehensive ratio analysis and Discounted Cash Flow (DCF) valuation
- Implemented financial models in Microsoft Excel and Python for real company data

MCB Historical Prices — Exploratory Analysis

- Conducted exploratory data analysis on MCB Bank historical stock price data
- Identified long-term trends, seasonality patterns, and notable price movements

Stock Price Simulation (MCB) — Monte Carlo

- Applied Monte Carlo simulation methods to model possible future stock price paths for MCB Bank
- Quantified risk and uncertainty across thousands of simulated price trajectories

MCB Return Distribution Analysis

- Computed and visualized daily log returns for MCB Bank stock
- Produced statistical summary including distribution shape, skewness, and volatility metrics

Interactive Dashboards

- Built exploratory data analysis dashboards using Plotly Dash and Power BI

- Designed interactive visualizations enabling dynamic data exploration

ACHIEVEMENTS

BusinessEXPO Event Organizer

- Organized and coordinated the BusinessEXPO event, demonstrating leadership, planning, and execution skills in a competitive business-focused environment

Best Girls Representative Award

- Recognized as Best Girls Representative for outstanding dedication and advocacy within the student community

LANGUAGES

English (Proficient)